

Numerical Analysis-Final Exam 2023

Name _____

ERP.No _____

1 Instructions

- Attempt all questions.
- Please limit your solutions to the space given against each question.
- You are allowed to use simple scientific calculators.
- You may use answer book for rough work.
- It is an open-book and open-notes exam.
- Sharing of notes and books is strictly not allowed.

Instructor: Abdul Majid

1. Consider the equation $f(x) = 0$, which has a root in the interval $[0, b]$, where $b > 0$. Choose the maximum value of b which will require bisection method to reach accuracy of 10^{-3} in 3 iterations.

2. Find value of step size h in Three-Point Endpoint formula which minimizes the total(round-off and truncation) error.

3. Consider the equation $f(x) = e^x - x^2$. Construct an equation $x = g(x)$ so that fixed point iterative scheme converges quadratically. Also, perform two iterations of the fixed point method using g .

4. Use an appropriate Newton-Cotes formula to evaluate $\int_0^1 \frac{1}{x \ln x}$ with $n = 2$. Also, try to calculate same integral using Gauss quadrature with $n = 3$.

5. Show that the following quadrature is stable with respect to round off error.

$$\int_a^b f(x)dx = h\frac{9}{4}f(x_1) + h\frac{3}{4}f(x_2),$$

where $h = \frac{b-a}{3}$, $x_0 = a$, $x_1 = a + h$ and $x_2 = b$.

6. Use an optimal value of the step size h in the best available ThreePoint derivative formula to find the derivative of $f(x) = x - e^x$ at $x = 0.1$. Also, calculate the approximate error.

7. Use the optimal value of step size h in the composite $3/8th$ Simpson's rule and evaluate the integral $\int_1^2 (x - \ln x) dx$ using $n = 8$.

8. Derive the composite integral formula of open Newton-Cotes for $n = 1$.

9. Show that the following differential equations has a unique solution, and solve it numerically using Euler's method with $n = 10$ grid points.

$$\frac{dy}{dt} = y^2 \sin t, \text{ where } y(0) = 1 \text{ and } t \in [-\infty, \infty], y \in [-1, 1].$$

10. Perform three iterations of Gauss-Seidel method to solve the following linear system. Also, calculate 2-norm of the difference of last two iterations.

$$\begin{aligned}4x - y &= 11 \\-x + 4y - z &= -15 \\-y + 4z &= 20.\end{aligned}$$